Circuit Simulation Lab: 4
Encoders

Introduction

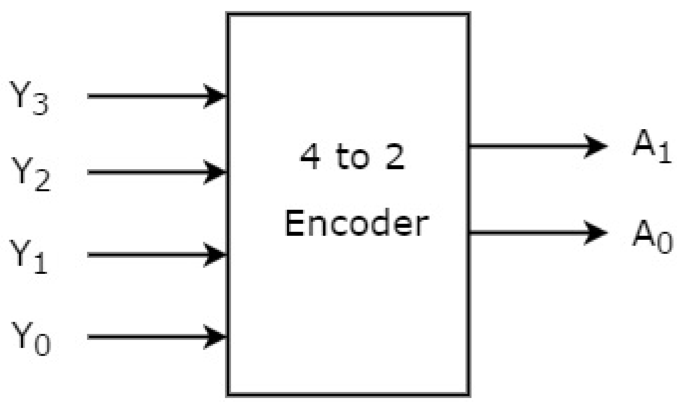
The purpose of this experiment is to introduce the design of simple combinational circuits, in this case Encoders. An Encoder is a combinational circuit that performs the reverse operation of Decoder. It has maximum of 2n input lines and 'n' output lines. It will produce a binary code equivalent to the input, which is active High. Therefore, the encoder encodes 2n input lines with 'n' bits. It is optional to represent the enable signal in encoders.

Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

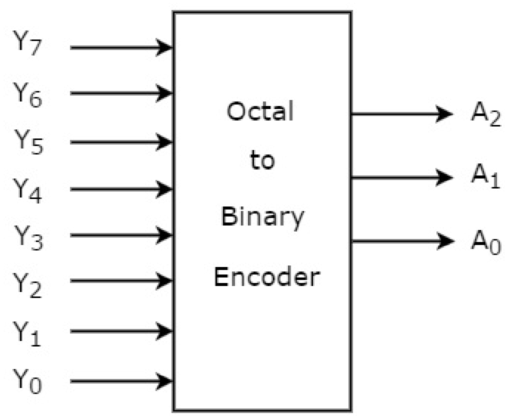
Logic Diagram



4 to 2 Encoder

Inputs				Outputs	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Truth Table



Octal to Binary Encoder

Describe the above circuits in Verilog HDL and capture the Waveforms

1. ***Dataflow modeling***
2. ***Behavior modeling***
3. ***Structural modeling***

Questions to answered after this lab

1. *Implement full adder by using suitable decoder.*
2. *Write short notes on "test bench" with examples.*